PROCEDURE STEP NO: 1

TIMEBASE ACCURACY

Conditions:

Reference: Internal.

NOMINAL VALUE (MHz)	*LOWER LIMIT dev(Hz)*	*BEFORE ADJ. dev(Hz)*	*AFTER ADJ. dev(Hz)*	*UPPER LIMIT dev(Hz)*	*UNCERTAINTY ±(Hz)*
10	-100	23		100	29

PROCEDURE STEP NO: 2

INPUT RESISTANCE DC

Conditions:

Input Resistance: 1 MOhm

Channel 1

VERTICAL SCALE (V/div)	*LOWER LIMIT (MOhm)*	*BEFORE ADJ. (MOhm)*	*AFTER ADJ. (MOhm)*	*UPPER LIMIT (MOhm)*	*UNCERTAINTY ±(MOhm)*
0.01	0.9900	0.9965		1.0100	0.002
1	0.9900	0.9981		1.0100	0.002

Channel 2

VERTICAL SCALE (V/div)	*LOWER LIMIT (MOhm)*	*BEFORE ADJ. (MOhm)*	*AFTER ADJ. (MOhm)*	*UPPER LIMIT (MOhm)*	*UNCERTAINTY ±(MOhm)*
0.01	0.9900	0.9977		1.0100	0.002
1	0.9900	0.9974		1.0100	0.002

Channel 3

VERTICAL SCALE (V/div)	*LOWER LIMIT (MOhm)*	*BEFORE ADJ. (MOhm)*	*AFTER ADJ. (MOhm)*	*UPPER LIMIT (MOhm)*	*UNCERTAINTY ±(MOhm)*
0.01	0.9900	0.9973		1.0100	0.002
1	0.9900	0.9974		1.0100	0.002

Channel 4

VERTICAL SCALE (V/div)	*LOWER LIMIT (MOhm)*	*BEFORE ADJ. (MOhm)*	*AFTER ADJ. (MOhm)*	*UPPER LIMIT (MOhm)*	*UNCERTAINTY ±(MOhm)*
0.01	0.9900	0.9972		1.0100	0.002
1	0.9900	0.9976		1.0100	0.002

PROCEDURE STEP NO: 3.1

DC GAIN ACCURACY

Channel 1

VERT SCALE (V/div)	*CAL VOLTAGE (V)*	*LOWER LIMIT dev(%)*	*BEFORE ADJ. dev(%)*	*AFTER ADJ. dev(%)*	*UPPER LIMIT dev(%)*	*UNCERTAINTY ±(%)*
100	±300	-1.5	-0.82		1.5	0.25
50	±150	-1.5	-0.86		1.5	0.25
40	±120	-1.5	-0.88		1.5	0.25
20	±60	-1.5	-0.88		1.5	0.25
10	±30	-1.5	-0.67		1.5	0.25
5	±15	-1.5	-0.65		1.5	0.25
4	±12	-1.5	-0.91		1.5	0.25
2	±6	-1.5	-0.89		1.5	0.25
1	±3	-1.5	-0.91		1.5	0.25
0.5	±1.5	-1.5	-0.82		1.5	0.25
0.4	±1.2	-1.5	-0.84		1.5	0.25
0.2	±0.6	-1.5	-0.74		1.5	0.25
0.1	±0.3	-1.5	-0.64		1.5	0.25
0.05	±0.15	-1.5	-0.88		1.5	0.25
0.04	±0.12	-1.5	-0.70		1.5	0.25
0.02	±0.06	-1.5	-0.81		1.5	0.25
0.01	±0.03	-1.5	-0.93		1.5	0.25
0.005	±0.015	-2.0	-1.13		2.0	0.25
0.004	±0.012	-2.0	-1.10		2.0	0.25
0.002	±0.004	-2.5	-1.30		2.5	0.25

PROCEDURE STEP NO: 3.2

DC GAIN ACCURACY

Channel 2

VERT SCALE (V/div)	*CAL VOLTAGE (V)*	*LOWER LIMIT dev(%)*	*BEFORE ADJ. dev(%)*	*AFTER ADJ. dev(%)*	*UPPER LIMIT dev(%)*	*UNCERTAINTY ±(%)*
100	±300	-1.5	-0.24		1.5	0.25
50	±150	-1.5	-0.35		1.5	0.25
40	±120	-1.5	-0.37		1.5	0.25
20	±60	-1.5	-0.47		1.5	0.25
10	±30	-1.5	-0.28		1.5	0.25
5	±15	-1.5	-0.33		1.5	0.25
4	±12	-1.5	-0.48		1.5	0.25
2	±6	-1.5	-0.48		1.5	0.25
1	±3	-1.5	-0.62		1.5	0.25
0.5	±1.5	-1.5	-0.58		1.5	0.25
0.4	±1.2	-1.5	-0.24		1.5	0.25
0.2	±0.6	-1.5	-0.53		1.5	0.25
0.1	±0.3	-1.5	-0.38		1.5	0.25
0.05	±0.15	-1.5	-0.52		1.5	0.25
0.04	±0.12	-1.5	-0.53		1.5	0.25
0.02	±0.06	-1.5	-0.57		1.5	0.25
0.01	±0.03	-1.5	-0.76		1.5	0.25
0.005	±0.015	-2.0	-0.91		2.0	0.25
0.004	±0.012	-2.0	-1.00		2.0	0.25
0.002	±0.004	-2.5	-1.72		2.5	0.25

PROCEDURE STEP NO: 3.3

DC GAIN ACCURACY

Channel 3

VERT SCALE (V/div)	*CAL VOLTAGE (V)*	*LOWER LIMIT dev(%)*	*BEFORE ADJ. dev(%)*	*AFTER ADJ. dev(%)*	*UPPER LIMIT dev(%)*	*UNCERTAINTY ±(%)*
100	±300	-1.5	-0.80		1.5	0.25
50	±150	-1.5	-0.70		1.5	0.25
40	±120	-1.5	-0.67		1.5	0.25
20	±60	-1.5	-0.72		1.5	0.25
10	±30	-1.5	-0.51		1.5	0.25
5	±15	-1.5	-0.64		1.5	0.25
4	±12	-1.5	-0.74		1.5	0.25
2	±6	-1.5	-0.77		1.5	0.25
1	±3	-1.5	-0.64		1.5	0.25
0.5	±1.5	-1.5	-0.89		1.5	0.25
0.4	±1.2	-1.5	-0.62		1.5	0.25
0.2	±0.6	-1.5	-0.56		1.5	0.25
0.1	±0.3	-1.5	-0.79		1.5	0.25
0.05	±0.15	-1.5	-0.74		1.5	0.25
0.04	±0.12	-1.5	-0.81		1.5	0.25
0.02	±0.06	-1.5	-0.88		1.5	0.25
0.01	±0.03	-1.5	-0.95		1.5	0.25
0.005	±0.015	-2.0	-0.91		2.0	0.25
0.004	±0.012	-2.0	-1.25		2.0	0.25
0.002	±0.004	-2.5	-1.53		2.5	0.25

PROCEDURE STEP NO: 3.4

DC GAIN ACCURACY

Channel 4

VERT SCALE (V/div)	*CAL VOLTAGE (V)*	*LOWER LIMIT dev(%)*	*BEFORE ADJ. dev(%)*	*AFTER ADJ. dev(%)*	*UPPER LIMIT dev(%)*	*UNCERTAINTY ±(%)*
100	±300	-1.5	-0.59		1.5	0.25
50	±150	-1.5	-0.70		1.5	0.25
40	±120	-1.5	-0.66		1.5	0.25
20	±60	-1.5	-0.76		1.5	0.25
10	±30	-1.5	-0.40		1.5	0.25
5	±15	-1.5	-0.40		1.5	0.25
4	±12	-1.5	-0.55		1.5	0.25
2	±6	-1.5	-0.64		1.5	0.25
1	±3	-1.5	-0.61		1.5	0.25
0.5	±1.5	-1.5	-0.73		1.5	0.25
0.4	±1.2	-1.5	-0.58		1.5	0.25
0.2	±0.6	-1.5	-0.50		1.5	0.25
0.1	±0.3	-1.5	-0.53		1.5	0.25
0.05	±0.15	-1.5	-0.57		1.5	0.25
0.04	±0.12	-1.5	-0.75		1.5	0.25
0.02	±0.06	-1.5	-0.73		1.5	0.25
0.01	±0.03	-1.5	-0.77		1.5	0.25
0.005	±0.015	-2.0	-0.99		2.0	0.25
0.004	±0.012	-2.0	-1.27		2.0	0.25
0.002	±0.004	-2.5	-1.39		2.5	0.25

PROCEDURE STEP NO: 4.1

OFFSET ACCURACY

Conditions:

RTH Position set to 0 div

Channel 1

VERT SCALE *(V/div)*	*OFFSET & CAL VOLTAGE (V)*	*LOWER LIMIT dev(mV)*	*BEFORE ADJ. dev(mV)*	*AFTER ADJ. dev(mV)*	*UPPER LIMIT dev(mV)*	*UNCERTAINTY ±(mV)*
0.002	0	-1.7	-0.447		1.7	0.02
0.002	4	-21.7	-1.285		21.7	0.25
0.002	-4	-21.7	0.925		21.7	0.25
0.05	0	-6.5	-0.23		6.5	0.45
0.05	4	-26.5	-0.80		26.5	0.45
0.05	-4	-26.5	1.18		26.5	0.45
0.1	0	-11.5	-2.57		11.5	0.90
0.1	4	-31.5	-3.87		31.5	0.90
0.1	-4	-31.5	-1.21		31.5	0.90
0.5	0	-51.5	-1.3		51.5	4.5
0.5	4	-71.5	-3.1		71.5	4.5
0.5	-4	-71.5	-0.8		71.5	4.5
1	0	-101.5	-35.0		101.5	9.0
1	200	-1101.5	-8.0		1101.5	16.1
1	-200	-1101.5	-43.6		1101.5	16.1
10	0	-1002	-63		1002	90
10	200	-2002	-35		2002	91
10	-200	-2002	-81		2002	91
20	0	-2000	-60		2000	180
20	200	-3000	-100		3000	180
20	-200	-3000	-10		3000	180

PROCEDURE STEP NO: 4.2

OFFSET ACCURACY

Conditions:

RTH Position set to 0 div

Channel 2

VERT SCALE *(V/div)*	*OFFSET & CAL VOLTAGE (V)*	*LOWER LIMIT dev(mV)*	*BEFORE ADJ. dev(mV)*	*AFTER ADJ. dev(mV)*	*UPPER LIMIT dev(mV)*	*UNCERTAINTY ±(mV)*
0.002	0	-1.7	-0.216		1.7	0.02
0.002	4	-21.7	-0.060		21.7	0.25
0.002	-4	-21.7	0.020		21.7	0.25
0.05	0	-6.5	0.22		6.5	0.45
0.05	4	-26.5	0.31		26.5	0.45
0.05	-4	-26.5	0.41		26.5	0.45
0.1	0	-11.5	-1.30		11.5	0.90
0.1	4	-31.5	-1.32		31.5	0.90
0.1	-4	-31.5	-1.50		31.5	0.90
0.5	0	-51.5	-1.9		51.5	4.5
0.5	4	-71.5	-0.6		71.5	4.5
0.5	-4	-71.5	-2.5		71.5	4.5
1	0	-101.5	-9.1		101.5	9.0
1	200	-1101.5	40.7		1101.5	16.1
1	-200	-1101.5	-55.2		1101.5	16.1
10	0	-1002	4		1002	90
10	200	-2002	79		2002	91
10	-200	-2002	-33		2002	91
20	0	-2000	10		2000	180
20	200	-3000	240		3000	180
20	-200	-3000	-240		3000	180

PROCEDURE STEP NO: 4.3

OFFSET ACCURACY

Conditions:

RTH Position set to 0 div

Channel 3

VERT SCALE *(V/div)*	*OFFSET & CAL VOLTAGE (V)*	*LOWER LIMIT dev(mV)*	*BEFORE ADJ. dev(mV)*	*AFTER ADJ. dev(mV)*	*UPPER LIMIT dev(mV)*	*UNCERTAINTY ±(mV)*
0.002	0	-1.7	0.134		1.7	0.02
0.002	4	-21.7	-1.159		21.7	0.25
0.002	-4	-21.7	1.766		21.7	0.25
0.05	0	-6.5	0.17		6.5	0.45
0.05	4	-26.5	-1.37		26.5	0.45
0.05	-4	-26.5	1.76		26.5	0.45
0.1	0	-11.5	0.96		11.5	0.90
0.1	4	-31.5	-0.11		31.5	0.90
0.1	-4	-31.5	1.44		31.5	0.90
0.5	0	-51.5	-2.5		51.5	4.5
0.5	4	-71.5	-2.6		71.5	4.5
0.5	-4	-71.5	-2.5		71.5	4.5
1	0	-101.5	-9.3		101.5	9.0
1	200	-1101.5	24.3		1101.5	16.1
1	-200	-1101.5	-30.9		1101.5	16.1
10	0	-1002	-36		1002	90
10	200	-2002	-26		2002	91
10	-200	-2002	-100		2002	91
20	0	-2000	-80		2000	180
20	200	-3000	-40		3000	180
20	-200	-3000	-250		3000	180

PROCEDURE STEP NO: 4.4

OFFSET ACCURACY

Conditions:

RTH Position set to 0 div

Channel 4

VERT SCALE *(V/div)*	*OFFSET & CAL VOLTAGE (V)*	*LOWER LIMIT dev(mV)*	*BEFORE ADJ. dev(mV)*	*AFTER ADJ. dev(mV)*	*UPPER LIMIT dev(mV)*	*UNCERTAINTY ±(mV)*
0.002	0	-1.7	-0.109		1.7	0.02
0.002	4	-21.7	0.222		21.7	0.25
0.002	-4	-21.7	-0.315		21.7	0.25
0.05	0	-6.5	0.14		6.5	0.45
0.05	4	-26.5	0.26		26.5	0.45
0.05	-4	-26.5	-0.29		26.5	0.45
0.1	0	-11.5	-0.38		11.5	0.90
0.1	4	-31.5	-0.75		31.5	0.90
0.1	-4	-31.5	-0.19		31.5	0.90
0.5	0	-51.5	0.1		51.5	4.5
0.5	4	-71.5	-0.1		71.5	4.5
0.5	-4	-71.5	1.2		71.5	4.5
1	0	-101.5	-3.1		101.5	9.0
1	200	-1101.5	17.3		1101.5	16.1
1	-200	-1101.5	-17.2		1101.5	16.1
10	0	-1002	33		1002	90
10	200	-2002	68		2002	91
10	-200	-2002	7		2002	91
20	0	-2000	260		2000	180
20	200	-3000	40		3000	180
20	-200	-3000	280		3000	180

PROCEDURE STEP NO: 5.1

ANALOG BANDWIDTH – CHANNEL 1

Conditions:

Vertical Scale: 10 mV/div
Horizontal Scale: 5 ns/div up to 200 MHz and 1 ns/div at 350 MHz

TEST FREQUENCY (MHz)	*LOWER LIMIT dev(dB)*	*BEFORE ADJ. dev(dB)*	*AFTER ADJ. dev(dB)*	*UPPER LIMIT dev(dB)*	*UNCERTAINTY ±(dB)*
60	-3.0	0.13		~	0.20
100	-3.0	0.23		~	0.20
200	-3.0	0.72		~	0.30
350	-3.0	-0.72		~	0.50

Conditions:

Vertical Scale: 100 mV/div
Horizontal Scale: 5 ns/div up to 200 MHz and 1 ns/div at 350 MHz

TEST FREQUENCY (MHz)	*LOWER LIMIT dev(dB)*	*BEFORE ADJ. dev(dB)*	*AFTER ADJ. dev(dB)*	*UPPER LIMIT dev(dB)*	*UNCERTAINTY ±(dB)*
60	-3.0	0.24		~	0.20
100	-3.0	0.33		~	0.20
200	-3.0	0.67		~	0.30
350	-3.0	-0.85		~	0.50

Conditions:

Vertical Scale: 1 V/div
Horizontal Scale: 5 ns/div up to 200 MHz and 1 ns/div at 350 MHz

TEST FREQUENCY (MHz)	*LOWER LIMIT dev(dB)*	*BEFORE ADJ. dev(dB)*	*AFTER ADJ. dev(dB)*	*UPPER LIMIT dev(dB)*	*UNCERTAINTY ±(dB)*
60	-3.0	0.13		~	0.20
100	-3.0	0.47		~	0.20
200	-3.0	0.41		~	0.30
350	-3.0	-1.60		~	0.50

PROCEDURE STEP NO: 5.2

ANALOG BANDWIDTH – CHANNEL 2

Conditions:

Vertical Scale: 10 mV/div
Horizontal Scale: 5 ns/div up to 200 MHz and 1 ns/div at 350 MHz

TEST FREQUENCY (MHz)	*LOWER LIMIT dev(dB)*	*BEFORE ADJ. dev(dB)*	*AFTER ADJ. dev(dB)*	*UPPER LIMIT dev(dB)*	*UNCERTAINTY ±(dB)*
60	-3.0	0.31		~	0.20
100	-3.0	0.34		~	0.20
200	-3.0	0.73		~	0.30
350	-3.0	-0.59		~	0.50

Conditions:

Vertical Scale: 100 mV/div
Horizontal Scale: 5 ns/div up to 200 MHz and 1 ns/div at 350 MHz

TEST FREQUENCY (MHz)	*LOWER LIMIT dev(dB)*	*BEFORE ADJ. dev(dB)*	*AFTER ADJ. dev(dB)*	*UPPER LIMIT dev(dB)*	*UNCERTAINTY ±(dB)*
60	-3.0	0.12		~	0.20
100	-3.0	0.48		~	0.20
200	-3.0	0.83		~	0.30
350	-3.0	-0.40		~	0.50

Conditions:

Vertical Scale: 1 V/div
Horizontal Scale: 5 ns/div up to 200 MHz and 1 ns/div at 350 MHz

TEST FREQUENCY (MHz)	*LOWER LIMIT dev(dB)*	*BEFORE ADJ. dev(dB)*	*AFTER ADJ. dev(dB)*	*UPPER LIMIT dev(dB)*	*UNCERTAINTY ±(dB)*
60	-3.0	0.13		~	0.20
100	-3.0	0.56		~	0.20
200	-3.0	0.71		~	0.30
350	-3.0	-1.32		~	0.50

PROCEDURE STEP NO: 5.3

ANALOG BANDWIDTH – CHANNEL 3

Conditions:

Vertical Scale: 10 mV/div
Horizontal Scale: 5 ns/div up to 200 MHz and 1 ns/div at 350 MHz

TEST FREQUENCY (MHz)	*LOWER LIMIT dev(dB)*	*BEFORE ADJ. dev(dB)*	*AFTER ADJ. dev(dB)*	*UPPER LIMIT dev(dB)*	*UNCERTAINTY ±(dB)*
60	-3.0	0.17		~	0.20
100	-3.0	0.44		~	0.20
200	-3.0	0.71		~	0.30
350	-3.0	-0.63		~	0.50

Conditions:

Vertical Scale: 100 mV/div
Horizontal Scale: 5 ns/div up to 200 MHz and 1 ns/div at 350 MHz

TEST FREQUENCY (MHz)	*LOWER LIMIT dev(dB)*	*BEFORE ADJ. dev(dB)*	*AFTER ADJ. dev(dB)*	*UPPER LIMIT dev(dB)*	*UNCERTAINTY ±(dB)*
60	-3.0	0.16		~	0.20
100	-3.0	0.49		~	0.20
200	-3.0	0.88		~	0.30
350	-3.0	-0.33		~	0.50

Conditions:

Vertical Scale: 1 V/div
Horizontal Scale: 5 ns/div up to 200 MHz and 1 ns/div at 350 MHz

TEST FREQUENCY (MHz)	*LOWER LIMIT dev(dB)*	*BEFORE ADJ. dev(dB)*	*AFTER ADJ. dev(dB)*	*UPPER LIMIT dev(dB)*	*UNCERTAINTY ±(dB)*
60	-3.0	0.23		~	0.20
100	-3.0	0.49		~	0.20
200	-3.0	0.88		~	0.30
350	-3.0	-1.42		~	0.50

PROCEDURE STEP NO: 5.4

ANALOG BANDWIDTH – CHANNEL 4

Conditions:

Vertical Scale: 10 mV/div
Horizontal Scale: 5 ns/div

TEST FREQUENCY (MHz)	*LOWER LIMIT dev(dB)*	*BEFORE ADJ. dev(dB)*	*AFTER ADJ. dev(dB)*	*UPPER LIMIT dev(dB)*	*UNCERTAINTY ±(dB)*
60	-3.0	0.32		~	0.20
100	-3.0	0.60		~	0.20
200	-3.0	1.00		~	0.30
350	-3.0	-0.36		~	0.50

Conditions:

Vertical Scale: 100 mV/div
Horizontal Scale: 5 ns/div

TEST FREQUENCY (MHz)	*LOWER LIMIT dev(dB)*	*BEFORE ADJ. dev(dB)*	*AFTER ADJ. dev(dB)*	*UPPER LIMIT dev(dB)*	*UNCERTAINTY ±(dB)*
60	-3.0	0.18		~	0.20
100	-3.0	0.49		~	0.20
200	-3.0	0.99		~	0.30
350	-3.0	-0.28		~	0.50

Conditions:

Vertical Scale: 1 V/div
Horizontal Scale: 5 ns/div

TEST FREQUENCY (MHz)	*LOWER LIMIT dev(dB)*	*BEFORE ADJ. dev(dB)*	*AFTER ADJ. dev(dB)*	*UPPER LIMIT dev(dB)*	*UNCERTAINTY ±(dB)*
60	-3.0	0.25		~	0.20
100	-3.0	0.61		~	0.20
200	-3.0	0.86		~	0.30
350	-3.0	-1.21		~	0.50

ROHDE&SCHWARZ

Calibration Certificate
Kalibrierschein

Certificate No. Zertifikatsnr. NATA 7900-316001200

Unit Data

Item Gegenstand	RTH1004 OSCILLOSCOPE 4CH
Manufacturer Hersteller	ROHDE & SCHWARZ
Type Typ	RTH1004
Material No. Materialnr.	1317.5000K04
Serial No. Seriennr.	104565

Order Data

Customer Auftraggeber	Tritium Pty Ltd 48 MILLER STREET MURARRIE QLD 4172
Order No. Bestellnummer	POAU-0024900
Date of Receipt Eingangsdatum	2021-09-20

Performance

Place and Date of Calibration Ort/Datum der Kalibrierung	R&S (AUSTRALIA) PTY. LTD. , 2021-09-23
Scope of Calibration Umfang der Kalibrierung	Standard Calibration
Statement of Compliance (Incoming) Konformitätsaussage (Anlieferung)	Defective
Statement of Compliance (Outgoing) Konformitätsaussage (Auslieferung)	All measured values are within the data sheet specifications.
Extent of Calibration Umfang des Kalibrierdokuments	2 Pages Calibration Certificate 13 Pages Outgoing Results

This calibration certificate documents the traceability to NMI (Australia), PTB (Germany), NIST (USA) or METAS (Switzerland), which realize the units of measurement according to the International System of Units (SI).
The expanded measurement uncertainty corresponds to the measurement results from the standard measurement uncertainty multiplied by the coverage factor k = 2. The true value is located in the corresponding interval with a probability of 95%.
Principles and methods of calibration comply with the technical requirements of ISO/IEC 17025.
NATA is a signatory to the multinational agreements of the International Laboratory Accreditation Cooperation (ILAC) for the mutual recognition of calibration certificates.
This calibration certificate may not be reproduced other than in full. Calibration certificates without signatures are not valid.
The user is obliged to have the object recalibrated at appropriate intervals.

NATA
ACCREDITED FOR
TECHNICAL COMPETENCE
Accredited for compliance with ISO/IEC 17025 - Calibration.
Accreditation Number: 3781
ilac-MRA

Date of issue (YYYY-MM-DD) Ausstellungsdatum	Authorised Signatory - Head of laboratory Zeichnungsberechtigte - Laborleitung	Calibration Officer Kalibrierbeauftragter
2021-09-27	Jing Wang	Nicolae Fieraru

ROHDE & SCHWARZ (Australia) Pty. Ltd.
ABN: 78 002 328 449, Unit 2, 75 Epping Road, North Ryde, NSW 2113,
Tel: +612 8874 5188 Fax: +612 8874 5198

Calibration Method
Kalibrieranweisung

Refer to calibration method and calibration instructions below.

Relative Humidity
Relative Luftfeuchte

15 - 80 %

Ambient Temperature
Umgebungstemperatur

23 ± 3 °C

Working standards used (having a significant effect on the accuracy)
Verwendete Gebrauchsnormale (mit signifikantem Einfluss auf die Genauigkeit)

Item	Type	Serial No.	Calibration Certificate No.	Cal. Due
GG XSRB RUBIDIUM-OSZILLATOR	XSRB	314725/009	RN210742	2022-05-05
34401A MULTIMETER	34401A	US36128743	1-14245602305-1	2023-03-30
5522A CALIBRATOR	5522A	3737902	1-13247405084-1	2021-09-29

Notes
Anmerkungen

Ancillary Functional Measurements:
The calibration procedure include ancillary functional measurements thataffect the traceability or measurement uncertainty of the calibration results. These functional measurement results are marked with an alphabetic character and are not intended to be used in support of the further dissemination of metrological traceability.

Comments on Measurement Results:
The measurement results in the test report stated below have been testedfor compliance with the given specifications and marked if necessary. The associated uncertainty of measurement has been taken into Measurement results that are not covered by the NATA accreditation are marked with 1.

UGB:
Uncertainty Guard Band - Measurement uncertainty violates the datasheet limit.

UGB1:
A compliance statement may be possible where a confidence level of less than 95 % is acceptable.
Die Bestätigung der Konformität ist möglich, sofern ein Grad des Vertrauens von weniger als 95% akzeptabel ist.

UGB2:
A non-compliance statement may be possible where a confidence level of less than 95 % is acceptable.
Die Bestätigung der Nicht-Konformität ist möglich, sofern ein Grad des Vertrauens von weniger als 95% akzeptabel ist.

Conformity statements take the measurement uncertainties into account.
Die Konformitätsaussagen berücksichtigen die Messunsicherheiten.

Ref: ILAC-G8:03/2009 'Guidelines on the Reporting of Compliance with Specification'

Calibration Method:
The instrument was tested in accordance with the relevant manual for Oscilloscope RTH1004, of Rohde & Schwarz and the in-house methods: RF-Z2, RF-Z51, DCVRES-2, FEC-001 contained in the Quality Assurance Manual.

Calibration Instructions:
The calibration of the instrument can be classified as a direct measurement. DC performance was tested by applying known voltages to instrument. Bandwidth was measured with a power standard. Frequency accuracy was measured with a frequency counter synchronized to a rubidium reference. The traceability is represented in the table Working Standards used.